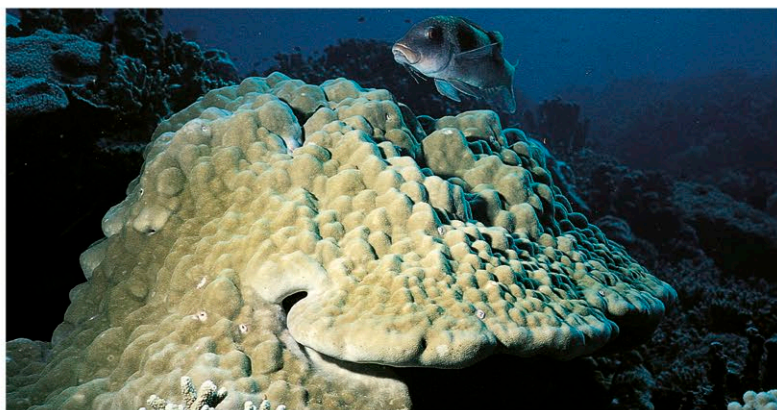


**PORITES SP.**

These massive corals, found in tropical waters, are second only to the staghorns as reef builders. They grow in the form of mounds, sometimes with large, blunt, fingerlike projections. Some Porites domes on the Great Barrier Reef are almost 33 ft (10 m) in diameter, and may be 200 years old. The individual polyps, however, are only about  $\frac{1}{8}$  in (3 mm) across.

**MUSHROOM CORALS**

While most corals are colonial, mushroom corals consist of a single polyp usually less than 20 in (50 cm) across. They live in tropical waters and, when young, are attached to the seabed by a stalk. When adult, as in the *Fungia* species above, they break away from the stalk and rest on the bottom.

**PAVONA SP.**

This species in the genus *Pavona* has a diameter of 10 in (25 cm) and is a common and conspicuous member of reefs from the Red Sea to the eastern Pacific. It has fine, starlike markings on the surface, and grows in spreading colonies that form columns, plates, or fronds.

**BRAIN CORALS**

The individual polyps of this *Lobophyllia* colony are relatively large— $1\frac{1}{4}$  in (3 cm) in diameter or more. They are arranged in meandering rows, giving the coral a brainlike form. Mature colonies of brain coral may weigh over a ton. Many species inhabit the Pacific Ocean.

**STAGHORN CORALS**

The staghorns, belonging to the genus *Acropora*, are among the fastest growing corals on reefs. They would quickly swamp most other species if not for the fact that they are fragile and easily damaged in storms. Found in tropical seas, many staghorn colonies resemble antlers or bushes—like the branching species on the left, which may be 6½ ft (2 m) tall. Some are more spreading and resemble the flat-topped acacia trees on the East African savanna. The platelike growth of the species below—up to 9¾ ft (3 m) across—intercepts light for its symbiotic algae and shades out potential competitors beneath.

antlerlike growth

platelike shape

**Class Scyphozoa****Jellyfish**

**Occurrence** 234 spp. worldwide; in marine plankton, a few on seaweed or the seabed



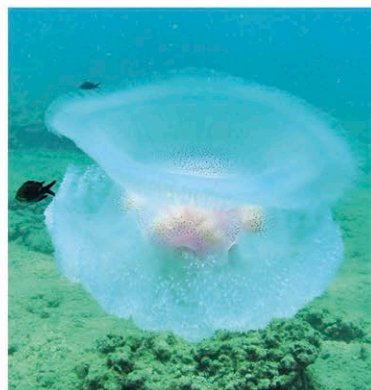
With bodies that can be over 6½ ft (2 m) across and tentacles that may extend many yards, jellyfish are among the largest animals found in plankton. A few species, however, attach themselves to seaweeds, and some spend much of their time on the seabed. A jellyfish's body, or "bell," is shaped like an inverted saucer or bowl, and consists mainly of a thick layer of jellylike mesoglea sandwiched between the

translucent, domed bell

**LION'S MANE JELLYFISH**

With a bell that can be over 6½ ft (2 m) across, the lion's mane jellyfish, *Cyanea capillata*, which lives in northern waters, is one of the biggest in the world. Its numerous long tentacles pose a danger to swimmers, and even when stranded on a beach, this jellyfish retains its ability to sting for a long time.

outer and inner cell layers (see p.537). The mouth in the center of the underside opens into a simple gut, and stinging tentacles hang down from the bell margin. These simple animals can have complex life cycles. Some (but not all) begin life as small polyps on the seabed. These split into a stack of tiny medusae, or baby jellyfish, which drift away to start life on their own, growing into sexually reproducing adults. Although jellyfish can swim by contracting their muscles, they make little headway against even a gentle current, and are often washed up on beaches after storms. Most are carnivorous, catching prey with their tentacles. This class includes species such as the moon jelly and lion's mane jellyfish.

**MARIVAGIA STELLATA**

Discovered in the Mediterranean in 2010, this species belongs to a group of jellyfish called rhizostomes—characterized by having a complex system of tiny mouths opening along specialized frilly "arms" that hang beneath the bell.

**MASTIGIAS PAPUA**

These broad-bodied jellyfish, some up to 7 in (19 cm) across, are most common in the tropics. They are rhizostomes and—like *Marivagia stellata* (left)—lack the margin of tentacles found around the bell of many other jellyfish.

**CASSIOPEA XAMACHANA**

A roughly disk-shaped jellyfish, 12 in (30 cm) across, this species lives in warm, shallow waters, and usually lies quietly pulsating upside down on the seabed. For nutrition, it relies on the symbiotic photosynthetic algae that fill the bladderlike structures around its mouth.